HDSS-Net: A Novel Hierarchically Designed Network With Spherical Space Classifier for Ship Recognition in SAR Images

Yuanzhe Shang, *Graduate Student Member, IEEE*, Wei Pu, *Member, IEEE*, Congwen Wu, *Graduate Student Member, IEEE*, Danling Liao, Xiaowo Xu, *Member, IEEE*, Chenwei Wang, *Member, IEEE*, Yulin Huang, *Senior Member, IEEE*, Yin Zhang, *Member, IEEE*, Junjie Wu, *Member, IEEE*, Jianyu Yang, *Senior Member, IEEE*, and Jianqi Wu

***Abstract*— Ship recognition in synthetic aperture radar (SAR) images is essential for many applications in maritime surveillance tasks. Recently, convolutional neural network (CNN)-based methods tend to be the mainstream in SAR recognition. Though considerable developments have been achieved, there are still several challenging issues toward superior ship recognition performance: 1) ships have a large variance in size, making it difficult to recognize ships by using single-scale features of CNN. 2) The SAR ship's large aspect ratio presents an obvious geometric characteristic. However, standard convolution is limited by the fixed convolution kernel, which is less effective in processing elongated SAR ships. 3) Existing CNN classifiers with softmax loss are less powerful to deal with intraclass diversity and interclass similarity in SAR ships. In this article, we propose a task-specific hierarchically designed network with a spherical space classifier (HDSS-Net) to alleviate the above-mentioned issues. First, to realize SAR ship recognition with large size variation, a feature aggregation module (FAM) is designed to obtain a feature pyramid that has strong representational power at all scales. Second, a FeatureBoost module (FBM) is devised to provide rectangular receptive fields to refine the features generated by FAM. Finally, a novel spherical space classifier (SSC) is proposed to expand the interclass margin and compress the intraclass feature distribution by fully taking advantage of the property of spherical space. The experimental results on two benchmark datasets (OpenSARShip and FUSAR-Ship) jointly show that the proposed HDSS-Net performs better than classic CNN methods and novel SAR ship recognition CNN methods.**

***Index Terms*— Convolutional neural networks (CNNs), hierarchically designed network, ship recognition, spherical space classifier (SSC), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has become a significant device for many remote sensing applications owing to its unique advantages of all-day, all-weather, high-resolution, and wide mapping functioning [1]. In particular, SAR automatic target recognition (ATR) appears to be one of the most demanding challenges in SAR applications [2]. The ATR system can remove men from the process of target recognition, which is desirable since the system with a man is generally slow, unreliable, vulnerable, and may limit the performance of the overall mission in real situations [3].

SAR ship recognition is one of the most practical applications among all the applications of SAR ATR. It is a basic and crucial research field for maritime surveillance tasks, such as making comprehensive marine decisions [4] and planning sea surface traffic [5]. Traditional SAR ship recognition algorithms mainly use some simple feature extraction methods to distinguish different ship classes, such as geometric features, scattering features, and structural features [6], [7], [8], [9], [10], [11]. Although these traditional methods achieved excellent performance on some specific occasions, they often have time-consuming and labor-consuming manual feature design procedures, complex mathematical theories, and weak migration capacity, making it challenging to meet the demands of intelligent remote sensing processing.

As a representative algorithm of deep learning, convolutional neural networks (CNNs) [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26] use an end-to-end framework to achieve the feature extraction and recognition, which can effectively improve SAR ship recognition accuracy. Zeng et al. [27] designed a novel CNN method with hybrid channel feature loss that jointly utilizes the information contained in the polarized channels. He et al. [28] proposed a densely connected triplet CNN for ship recognition in medium-resolution SAR images. They used a deep metric learning scheme to expand the distance between different classes. Xu and Lang [29] proposed a novel distribution shift metric learning method, which improves the original Laplacian regularized metric learning by adding an interclass distribution shift regularization term. Zhang and Zhang [30] proposed a squeeze-and-excitation Laplacian pyramid network with dual-polarization feature fusion (SE-LPN-DPFF) to sufficiently utilize dual-polarization characteristics and multiresolution features of SAR ship. Zhang et al. [31] proposed a novel deep learning network with a histogram of oriented gradient (HOG) feature fusion (HOG-ShipCLSNet). HOG-ShipCLSNet uses features extracted by

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The authors are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: shangyuanzhe@std.uestc.edu.cn; puwei@uestc.edu.cn).

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both CNN and traditional methods, which can improve the SAR ship recognition performance effectively. To summarize, the CNN methods have become one of the mainstreams for SAR ship recognition.

Although the CNN methods for SAR ship recognition demonstrate superior performance compared with traditional methods, achieving high-accuracy recognition still poses some core challenges. Fig. 1 shows several challenging issues.

1) First, the same class of ships has a large variance in size, as shown in the third and fourth columns of Fig. 1. Take the container ship as an example, due to the different loadings, the length of the 10 000-ton container ship and the 160 000-ton container ship is five times different, and the width is three times different. Thus, it is difficult to achieve SAR ship recognition of various sizes by using a single-scale feature of CNN.
2) Second, the large aspect ratio is an obvious geometric characteristic of ships. Standard convolution is limited by the fixed convolution kernel, which is less effective in processing elongated SAR ships.
3) Third, owing to the complex imaging conditions and diversified looking angles between the ship targets and the SAR sensors, ships in SAR images appear in various aspects with respect to the azimuth direction and different scattering characteristics. In addition, as shown in the first and second columns of Fig. 1, ships in the same class can differ in structure, while ships in different classes would also have similar size, topside structure, and so on [28], [32]. Due to the impacts of these factors, intraclass diversity and interclass similarity are present. However, classic CNN classifiers with softmax loss[1] lacks the power of discrimination to deal with this issue.

In this article, we propose an HDSS-Net to address ship recognition issues in SAR images. The proposed HDSS-Net contains three phases from top to bottom. In the top phase, to realize SAR ship recognition with size variation, a feature aggregation module (FAM) is proposed to aggregate features of multilayers from CNN. In the middle phase, a novel FeatureBoost module (FBM) is proposed to refine the multiscale features generated by the top phase. FBM designs rectangular convolutions in four different directions, which can better tackle the challenges posed by large aspect ratios and arbitrary orientations in SAR ship recognition. In the bottom phase, unlike classic classifiers, a novel spherical space classifier (SSC) is designed to address the intraclass diversity and interclass similarity in SAR ships. We systematically define the CNN-based SSC for the first time. The proposed SSC also combines the multihead self-attention (MHSA) and the Gaussian error linear unit (GeLU) activate function for making better recognition decisions. The main contributions of this article are as follows.

1) A task-specific HDSS-Net is proposed for SAR ship recognition. Based on the feature pyramid networks (FPNs) structure, an improved FAM is proposed to suit the SAR ship recognition task. By combining different scale features of CNN, FAM can create a feature pyramid that has strong representational power at all scales.

[1]Following [33], we define the softmax loss as the combination of the softmax function and cross entropy (CE) loss.

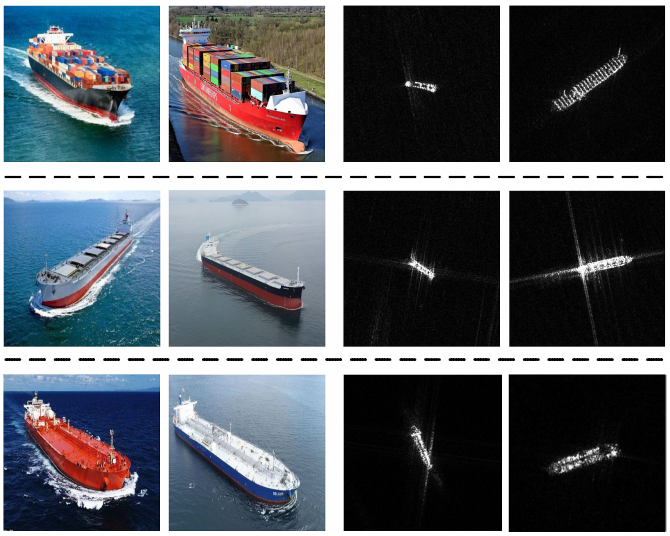


Fig. 1. Optical images and the corresponding SAR images of (Top) container ships, (Middle) bulk carriers, and (Bottom) tankers. First, one can find that in SAR images, the same class of ships has a large variance in size. Second, the large aspect ratio is an obvious geometric characteristic of ships. Third, large within-class diversity and between-class similarity exist, which will severely degenerate the recognition performance.

2) Previous methods extract ship features with square convolution kernel. However, the ship class with strip shape and arbitrary orientations cannot be effectively extracted by traditional square convolution kernel for its special shape prior characteristics. To suppress this problem, we proposed a novel FBM, which can integrate the ship shape into the end-to-end network. By introducing rectangular convolutions in four different directions (i.e., horizontal, vertical, main diagonal, and subdiagonal), FBM can encounter the limitation with square convolution and develop more discriminative representations.

3) A novel CNN-based SSC is proposed for the SAR ship recognition task. Of particular reference, the innovation and contribution of the CNN-based SSC contains the following.

1) We systematically define the CNN-based SSC for the first time. Compared with existing classifiers [28], [31], [34], [35], our SSC not only uses L_2 normalization to project the features onto the spherical space but also defines linear transform, logistic regression, MHSA, GeLU activate function, and other operations on the spherical space under the CNN architecture. Based on the theoretical framework of the proposed SSC, we can leverage the operations defined in the spherical space earlier to fully utilize the property of the spherical space, in order to achieve the purpose of expanding the interclass margin and compressing the intraclass feature distribution.
2) We combine the SSC with MHSA and GeLU activate function for the first time and apply them to the SAR ship recognition task. The MHSA can learn global feature dependencies and realize long-distance feature correlation calculation, enabling the incorporation of long-range feature dependencies into recognition

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decisions. The GeLU activate function can increase the nonlinear relationship between layers. It is more sensitive to varieties of the input features, enabling better convey subtle differences of the input features.

The rest of this article is arranged as follows. Section II presents HDSS-Net in detail and discusses why SSC with softmax loss owns a better performance than the classic classifier with softmax. Section III validates the effectiveness of the proposed method through extensive experiments. Finally, Section IV provides the conclusion.

II. Methodology

In this section, in order to realize effective recognition of SAR ships, starting from the characteristics of ship targets, HDSS-Net is constructed. We first describe the overall framework of HDSS-Net. Then, we elaborate on the three main phases of the proposed CNN model, respectively. Finally, we discuss why SSC with softmax loss owns a better performance than the classic classifier with softmax loss.

A. Overall Framework

The overall architecture of our proposed network is shown in Fig. 2. The whole architecture is constructed in three phases from top to bottom. In the top phase (Section III-B), we propose FAM, which combines the multilayer features of CNN to create a feature pyramid that has representational power at all scales. FAM helps the handling of SAR ships of diverse sizes. In the middle phase (Section III-C), considering the special shape prior characteristics of the ship class, FBM has been designed to enhance ship features, which incorporate the prior knowledge of the ship shape. In the bottom phase (Section III-D), we propose SSC. By fully taking advantage of the intrinsic structures of spherical space, SSC clusters the feature vectors from the same classes together and pulls apart those from different classes for effective SAR ship recognition. Combining MHSA and GeLU enables SSC to make better recognition decisions.

B. Framework in the Top Phase: Feature Extraction Backbone

Most CNN methods for SAR ship recognition only utilize features from the last layer. However, it is difficult to achieve SAR ship recognition of various sizes by using single-scale features. An effective way to solve this problem is to utilize the multiscale features of middle layers from CNN, such as FPN [36]. However, the FPN structure is designed for the detection task, so some small improvements need to be made. FAM is constructed for SAR ship recognition tasks based on the FPN structure. FAM combines different scale features of CNN to create a feature pyramid that has strong representational power at all scales.

ResNet-50 [16] is used as the backbone and divided into five convolution stages. Each stage outputs feature maps of different scales. The outputs of the last three stages are indicated as $\{C_3, C_4, C_5\}$. Low-level features C_3 contain more details and less semantics. On the contrary, C_4 and C_5 are high-level features with fewer details and more semantics. FAM aggregates C_3–C_5 feature layers and integrates semantic and detailed features to enhance the feature representation ability at all scales. These networks are shown in Fig. 3. By FAM, a feature pyramid that contains three groups of enhanced feature maps P_3, P_4, and P_5 can be obtained. The overall process of FAM is summarized as

$$I_l = \begin{cases} \mathrm{Conv}_{1\times1}(\mathrm{C}_5), & l = 5 \\ \mathrm{Upsample}(I_{l+1}) + \mathrm{Conv}_{1\times1}(\mathrm{C}_1), & l = 3, 4 \end{cases} \tag{1}$$

$$P_l = \begin{cases} \mathrm{Conv}_{3\times3}(I_l), & l = 5 \\ \mathrm{Interpolation}(\mathrm{Conv}_{3\times3}(I_l)), & l = 3, 4 \end{cases} \tag{2}$$

$$P = P_3 © P_4 © P_5 \tag{3}$$

where $I_l \in \mathbb{R}^{(224/2^l)\times(224/2^l)\times C}$, and $\mathrm{Conv}_{1\times1}$ and $\mathrm{Conv}_{3\times3}$ denote the 1×1 and 3×3 convolution, respectively. Upsample is the nearest neighbor upsampling with the factor of 2. $\mathrm{Conv}_{1\times1}$ is attached to $C_l (l = 3, 4, 5)$ to reduce the dimensions of I_l channel to 256. $\mathrm{Conv}_{3\times3}$ is attached to each fused feature map I_l to alleviate aliasing effect of upsampling [37]. It should be noted that we resize P_3 and P_4 to the same size as P_5 with interpolation strategies. Interpolation is the nearest neighbor interpolation algorithm, and © is the operation of concatenation. Finally, we concatenate feature pyramid P_3, P_4, and P_5 to get P. P has strong semantics at all scales for effective multiscale SAR ship recognition. To further refine the multiscale feature P, we develop FBM. The details of FBM are described as follows.

C. Architecture in the Middle Phase: FeatureBoost Module

Compared with other types of features, the ship class in SAR image has special shape prior characteristics. However, the ship class with strip shape and arbitrary orientations cannot be effectively extracted by standard square convolution kernel for its special shape prior characteristics. To improve the representations of the extracted features, FBM is proposed. FBM is developed to improve the performance of elongated SAR ship recognition by solving the issue of the fixed receptive field of the standard convolution kernel.

Standard convolutional layers generally employ square kernels, such as $K \times K$. Assuming the convolutional layer takes a C-channel feature map as input. Given the features $P \in \mathbb{R}^{W\times H\times C}$ and kernels $S \in \mathbb{R}^{K\times K\times C}$, the output feature map F_o can be denoted as

$$F_o = P * S \tag{4}$$

where * is the 2-D convolution operator. The standard square $K \times K$ kernels are not effective enough in the challenging SAR ship scenarios.

To handle the limitation encountered by the square convolution, we combine horizontal and vertical rectangular kernels [38], [39] with two innovative diagonal rectangular convolution kernels to fit the shape of the SAR ship. FBM combines a standard square convolution kernel and four rectangular convolution kernels. The original features are preserved by the square kernel, while horizontal, vertical, and diagonal kernels improve the details by providing rectangular receptive fields.

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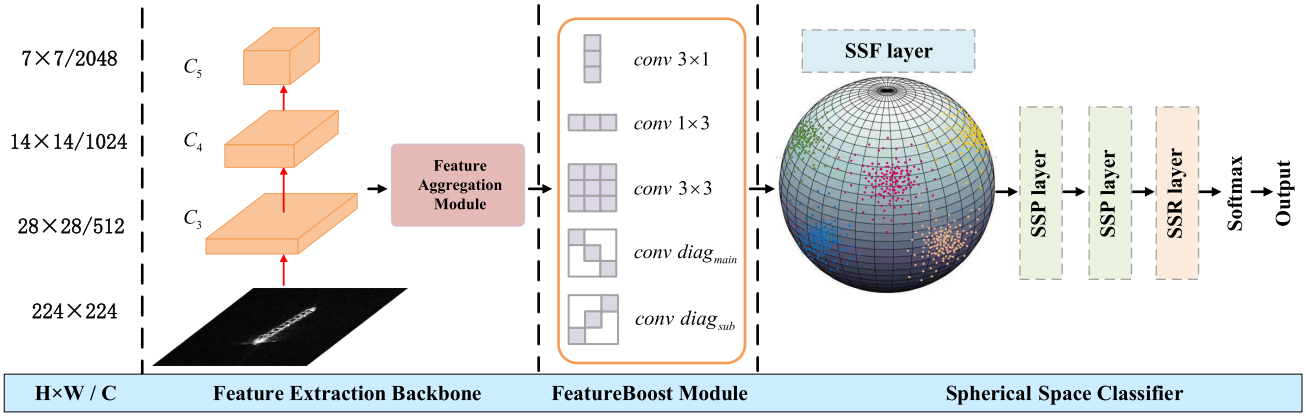


Fig. 2. Overall architecture of HDSS-Net. In the top phase, the features of an input SAR image are extracted to create a feature pyramid that has representational power at multiscales. In the middle phase, FBM is used to refine the features by providing rectangular receptive fields. In the bottom phase, the refined features are fed into SSC to recognize SAR ship targets.

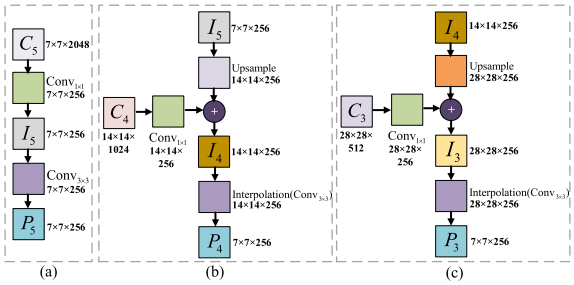


Fig. 3. Structure of FAM module. The feature maps of (a) P_5, (b) P_4, and (c) P_3 are attained. The $7 \times 7 \times 2048$ for C_5 means that there are 2048 feature maps with 7×7 pixels in C_5. This representation is also applicable to C_4, C_3, P_5, P_4, and P_3. The symbol "$\oplus$" means adding operation.

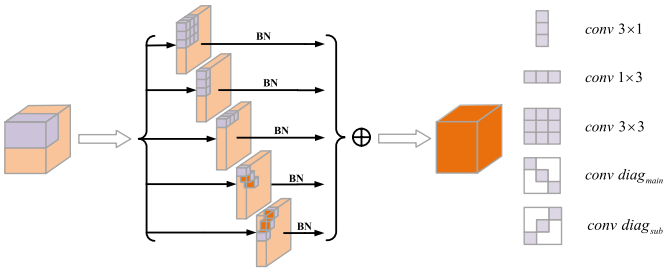


Fig. 4. Structure of FBM. In this figure, the FBM contains five parallel layers with different kernel sizes.

Fig. 4 illustrates the architecture of FBM. We construct a parallel organization of five branches with different kernel sizes. Concretely, for the replacement of a $K \times K$ square kernel $S \in \mathbb{R}^{K \times K \times C}$, FBM comprises five parallel branches, including $K \times K$, $1 \times K$, $K \times 1$, a main diagonal, and a subdiagonal convolution kernels, the parameters of the convolution kernels at these corresponding positions are added together as the output. Note that each branch here has applied BatchNormal. The horizontal kernels $S_1 \in \mathbb{R}^{1 \times K \times C}$, vertical kernels $S_2 \in \mathbb{R}^{K \times 1 \times C}$, main diagonal $S_3 \in \mathbb{R}^{[\text{diag}_{\text{main}}] \times C}$, and subdiagonal $S_4 \in \mathbb{R}^{[\text{diag}_{\text{sub}}] \times C}$ are rectangular kernels to fit the shape of SAR ships. The output of FBM can be described as

$$F = \text{ReLU}\left(P * S + \sum_{l=1}^{4} P * S_l\right) \tag{5}$$

where F and P indicate the output and input feature maps of FBM. S_l enhances the horizontally elongated kernels, vertically elongated kernels, and diagonally elongated kernels.

Small kernels are advantageous for obtaining the receptive fields of small ships, and the same receptive field for large ships can be obtained by using multiple convolutions with a small kernel. Hence, the kernel size parameter K is set to 3.

D. Classifier in the Bottom Phase: Spherical Space Classifier

This section introduces the details of how SSC is constructed. In the previous SAR ship recognition works, the classic classifiers are multilayer perceptrons (MLPs). Different from classic classifiers process features in the Euclidean space, SSC processes features in the non-Euclidean space. Based on the theoretical framework of the proposed SSC, we leverage the basic operations defined in the spherical space to fully utilize the property of the spherical space. A series of spherical space layers are constructed to achieve the recognition in spherical space. SSC also combines MHSA and GeLU activate functions for making better recognition decisions.

As shown in Fig. 5, the SSC consists of a spherical space feature (SSF) SSF layer, two spherical space perceptron (SSP) layers, and a spherical space logistic regression (SSR) layer. It contains a softmax function at the end. The SSF layer projects features onto the sphere to obtain spherical features. The SSP layer is utilized to perform a spherical space linear transform from one spherical space to another and increase the nonlinear relationship between spherical features. The SSR layer is designed for predicting recognition scores on the sphere. The construction of each layer is described in detail in the following.

1) SSF Layer: The first step of SSC is to obtain spherical features. For the convenience of description, we treat this operation as a component of SSC. Through the SSF layer, we normalize the extracted features to obtain features in spherical space with $\{f = r(F(x)/\|F(x)\|) : \|f\| = r\}$, where f is the feature vector and r is the radius of spherical space.

2) SSP Layer: The SSP layer is the extension of the perceptron layer of MLP from Euclidean space to spherical space. A perception layer of MLP consists of a linear transform and

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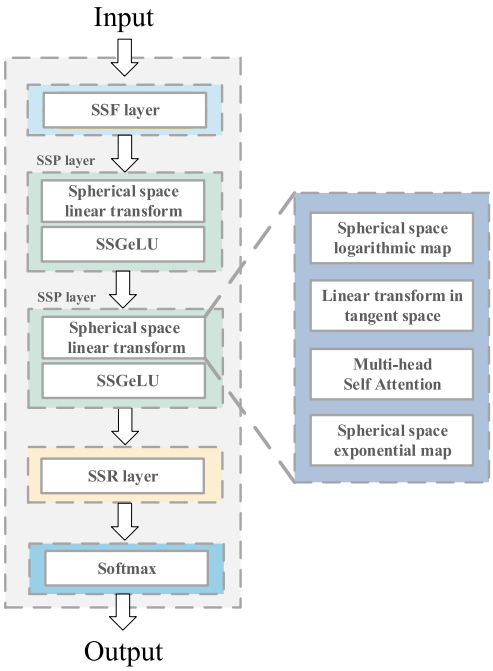


Fig. 5. Structure of SSC. It is constructed by an SSF transfer layer, two SSP layers, an SSR layer, and one more softmax function.

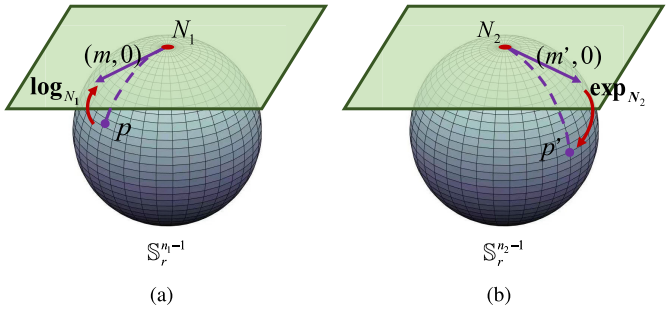


Fig. 6. (a) Spherical space logarithmic map. (b) Spherical space exponential map.

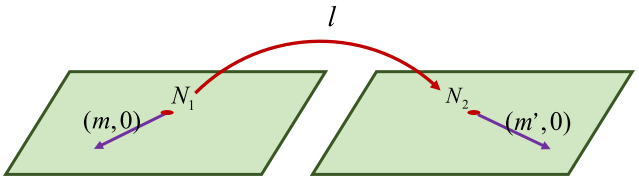


Fig. 7. Linear transform in the tangent space.

an activation function. Similarly, we define spherical linear transform and spherical activation functions for constructing the SSP layer.

Spherical Space Linear Transform: Similar to the design of hyperbolic neural network [40], the spherical linear transform is comprised of three elements: a spherical space logarithmic map, a linear transformation in tangent space, and a spherical space exponential map, which is illustrated in Fig. 5. Kirillov [41], Wilson and Hancock [42], and Gu [44] provide theoretical bases to achieve the spherical space linear transform. When conducting a spherical space linear transform from one spherical space to another, the process involves first projecting the features of the former spherical space onto its tangent space, which is essentially a hyperplane. Next, the projected features are transformed into the tangent space of the latter spherical space by the linear transform and perform MHSA. Finally, the transformed features are projected onto the latter spherical space by using the spherical exponential map. Mathematically, the spherical space linear transform l_s is defined by

$$l_s(p) = \exp_{N_2}((l_{\text{MHSA}}(\log_{N_1}(p)))) \tag{6}$$

where l_{MHSA} is a linear transform in tangent space with MHSA, $\exp_{N_2}$ and $\log_{N_1}$ are spherical space exponential and logarithmic maps, respectively. $N_i = (0, 0, \ldots, r)$ is the north pole, $i = 1, 2$. More details are described in the following.

Due to spherical space being a non-Euclidean space, it is difficult to establish linear transform. The study of spherical space has been observed to exhibit special local properties that enable simplification of computations by mapping to tangent space [43], [44]. Logarithmic and exponential maps are tools that allow us to connect the spherical space and its tangent space. The North Pole is an origin where a sphere touched the tangent space. Note that the notations do not imply the normal log and exp functions. As the tangent space is Euclidean [42], we can perform linear transform in a straight way.

Spherical Space Logarithmic and Exponential Maps: The logarithmic and exponential maps connect the tangent space and the spherical space. As shown in Fig. 6, define $N = (0, 0, \ldots, r) \in \mathbb{R}^n$ be the north pole of sphere $\mathbb{S}_r^{n-1} = \{p \in \mathbb{R}^n : \|p\| = r\}$, where sphere touched the tangent plane. N is the origin of the tangent space. The tangent space T_N at N becomes the hyperplane $e^T z - r = 0, \forall z \in \mathbb{R}^n$, where $e_n = (0, 0, \ldots, 1) \in \mathbb{R}^n$. Vector $\tilde{m}$ in T_N is defined as $\tilde{m} = (m, 0)$, where $m \in \mathbb{R}^{n-1}$ and the starting point of the vector $\tilde{m}$ is N. In addition, as shown in the purple line of Fig. 7, $\|\tilde{m}\|$ on T_N is the same as the geodesic distance between N and p on $\mathbb{S}_r^{n-1}$.

The logarithmic map $\log_N : \mathbb{S}_r^{n-1} \to T_N$ is given by

$$\log_N(p) = \frac{\varphi}{\sin\varphi}(p - N\cos\varphi) \quad \forall p \in \mathbb{S}_r^{n-1} \tag{7}$$

where $\varphi = \arccos(N^T p/r^2)$. The exponential map $\exp_N : T_N \to \mathbb{S}_r^{n-1}$ is calculated as

$$\exp_N(\tilde{m}) = N\cos\theta + \tilde{m}\frac{\sin\theta}{\theta} \quad \forall \tilde{m} \in T_N \tag{8}$$

where $\theta = (\|\tilde{m}\|/r)$.

Linear Transform in Tangent Space: To define the spherical space linear transform, we introduce spherical features on $\mathbb{S}_r^{n_1-1}$ to the tangent space T_{N_1} by $\log_{N_1}$, then transfer the conducted feature into T_{N_2} by linear transform l, and finally, the transferred features are sent to the sphere $\mathbb{S}_r^{n_2-1}$ by $\exp_{N_2}$, where $N_i = (0, 0, \ldots, r) \in \mathbb{R}^{n_i-1}$ is the north pole of $\mathbb{S}_r^{n_i-1}$, $i = 1, 2$. For any vector $\tilde{m}$ in T_{N_i}, the linear transform $l : T_{N_1} \to T_{N_2}$ can be expressed as

$$l(\tilde{m}, r) = (Wm + b, r) \quad \forall \tilde{m} \in T_{N_1} \tag{9}$$

where $W \in \mathbb{R}^{(n_2-1)*(n_1-1)}$ and $b \in \mathbb{R}^{n_2-1}$.

Multihead Self Attention: In addition, in the SSP layer, we embed MHSA after linear transform in tangent space.

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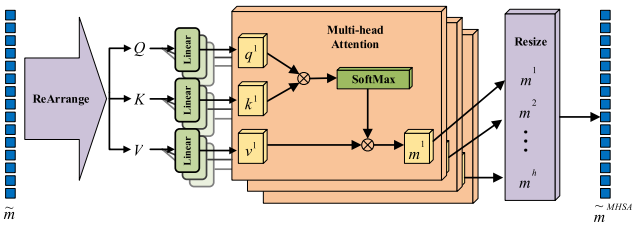


Fig. 8. Architecture of the MHSA.

The Transformer models are known for the long-distance modeling ability of MHSA [45], which is introduced to realize long-distance feature correlation calculation and learn global dependence among the features. MHSA can focus on the important features and suppress less significant features, enabling SSC to incorporate this long-range feature dependency into the recognition decisions to achieve effective enhancement of distinguishing SAR ship features.

As shown in Fig. 8, in MHSA, the given feature $\tilde{m} \in \mathbb{R}^D$ is rearranged and transformed into three sequences called Q, K, and V by the scale inner dot product with three learnable matrices W_q, W_k, and W_v as

$$\begin{aligned} Q &= \tilde{m} W_q \\ K &= \tilde{m} W_q \\ V &= \tilde{m} W_q \end{aligned} \tag{10}$$

where Q, K, and V indicate the query, key, and value for the attention process, respectively. After the transform, the resize calculation is applied to partition them into $[q^1, q^2, \ldots, q^h]$, $[k^1, k^2, \ldots, k^h]$, and $[v^1, v^2, \ldots, v^h]$, where h is the number of heads. The self-attention operation is proceeded in each head h. For a given embedded $\tilde{m}$, the attention head is formulated as

$$\tilde{m}^h(Q, K, V) = \sigma\left(\frac{Q^h (K^h)^T}{\sqrt{\text{Scale}}}\right) V^h \tag{11}$$

where σ denotes the softmax function. The output $[\tilde{m}^1, \tilde{m}^2, \ldots, \tilde{m}^h]$ is resized back to feature dimension D to remove the additional head dimension and $\tilde{m}^{\text{MHSA}} \in \mathbb{R}^D$ can be obtained.

Spherical Space Activation Function: As for the activation functions, the GeLU [46] is employed to increase the nonlinear relationship between the layers. The nonlinear activation function spherical space GeLU is formulated by

$$\text{SSGeLU}(x) = r\frac{\text{GeLU}(p)}{\|\text{GeLU}(p)\|}. \tag{12}$$

With the spherical space linear transform, MHSA, and spherical space activation function earlier, given input spherical feature f_{in} of the SSP layer, the output spherical feature f_{out} is calculated by

$$f_{\text{out}} = \text{SSGeLU}(l_s(f_{\text{in}})). \tag{13}$$

3) SSR Layer: This layer is designed for predicting recognition scores on the sphere. The recognition boundary of logistic regression is a circle in spherical space. The circle can be expressed as $w^T z + b = 0$, where $\|w\| = 1$, z is the spherical feature, and $b \in [-r, r]$.[2] Hence, we denote the SSR layer as

$$\delta_j = \text{Linear}(z_j, \|w_j\|, r \cdot \tanh(b_j)) \tag{14}$$

where δ_j is the jth output of SSR, z_j is the corresponding input of SSR, and linear$(\cdot)$ is the linear transformation function. $\delta_j = [\delta_1, \delta_2, \ldots, \delta_k]$ is a k dimension vector of logits for each sample, and k is the class number. We convert the logits to class probability by applying the softmax function. The probability of x being recognized in class j is formulated as

$$p(y = j|x) = \frac{\exp(\delta_j)}{\sum_{i=1}^{k} \exp \delta_i}. \tag{15}$$

Briefly, the basic workflow of HDSS-Net for SAR ship recognition is summarized by the pseudo codes in Algorithm 1 in the following.

Algorithm 1 HDSS-Net for SAR Ship Recognition.

Given: A gray SAR ship image $\mathbf{X} \in \mathbb{R}^{224\times224\times1}$;
Output: Recognition tensor $\mathbf{Out}_{HDSS-Net} \in \mathbb{R}^{nclass\times1}$.

Image $\mathbf{X}$ loading, hyperparamaters setting;
Phase1 :
    Extract hierarchical feature maps $\mathbf{C_3}$, $\mathbf{C_4}$, $\mathbf{C_5}$ with backbone;
    Build enhanced $\mathbf{P_3}$, $\mathbf{P_4}$, $\mathbf{P_5}$ with **FAM**;
    Concatenate $\mathbf{P_3}$, $\mathbf{P_4}$, $\mathbf{P_5}$ to get $\mathbf{P}$.
Phase2 :
    Input feature map $\mathbf{P}$ into **FBM**;
    Get improved feature map $\mathbf{F}$ by **FBM**;
    Flatten $\mathbf{F}$ by global average pooling (**GAP**).
Phase3 :
    Get spherical features $\mathbf{f} = r\frac{F(\mathbf{x})}{\|F(\mathbf{x})\|}$;
    Peform spherical space linear transform $l_s(\mathbf{f}) = \exp_{N_2}(l_{MHSA}(\log_{N_1}(\mathbf{f})))$;
    Peform spherical space activation function $\mathbf{f}_{out} = \text{SSGeLU}(l_s(\mathbf{f}))$;
    Peform spherical space linear transform $l_s(\mathbf{f}) = \exp_{N_2}(l_{MHSA}(\log_{N_1}(\mathbf{f}_{out})))$;
    Peform spherical space activation function $\mathbf{f}_{out} = \text{SSGeLU}(l_s(\mathbf{f}))$;
    Obtain the n-class dimension vector $\delta = \text{Linear}(\mathbf{f}_{out}, \|\mathbf{w}\|, r \cdot \tanh(\mathbf{b}))$;
    Predict the recognition scores $\mathbf{Out}_{HDSS-Net} = softmax(\delta)$.

E. Classic Classifier With Softmax Loss Versus SSC With Softmax Loss

As we mentioned earlier, SAR ship targets show large intraclass variability and small interclass separability. However, classic CNN classifiers with softmax lack the power of discrimination to deal with the issue. In contrast, SSC with softmax can show improved performance. This section will discuss the underlying reasons.

[2]The constraint that $b \in [-r, r]$ can be forced by modeling $b = r \cdot \tanh(b)$.

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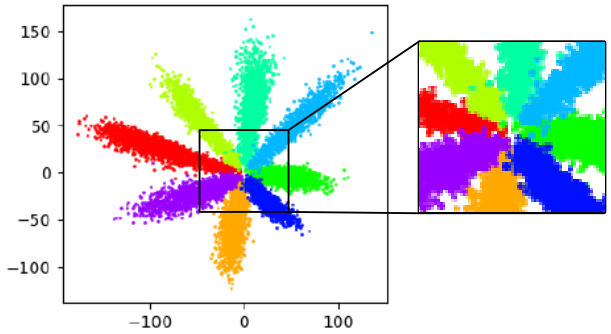
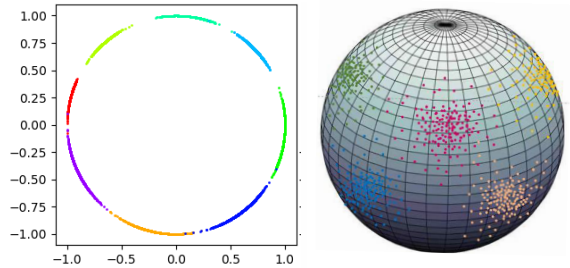


Fig. 9. Two-dimensional feature distribution using classic classifier with softmax loss. Each color represents a different class.



(a) (b)

Fig. 10. (a) Two-dimensional spherical feature distribution, in which the features are the same as Fig. 9. The feature vectors are enforced to locate onto a unit sphere. (b) Diagrammatic sketch of high dimension feature distribution after the SSF layer. The feature vectors are enforced to locate onto a unit hypersphere.

1) Classic Classifier With Softmax Loss: Most works use softmax loss to train the network. However, the softmax loss is insufficient to acquire the discriminating power for recognition. This part analyzes and answers why the classic classifier with softmax loss is insufficient to acquire the discriminating power for recognition.

In order to give an intuitive feeling about the softmax loss, a schematic Fig. 9 is given in which the number of the feature dimensions is reduced to 2-D for better visualization. Each dot represents a sample, i.e., an individual ship. Each color represents a class. As shown in Fig. 9, it can be seen that the feature distribution of the same class is "radial" when training with softmax loss. The variations in the radial direction lead to the intraclass diversity observed among the dots (individual samples) of the same class. Furthermore, the dots from different classes become intermingled around the zero point, leading to interclass similarity.

The reason why the softmax loss tends to create a "radial" feature distribution is that softmax loss always encourages features to have a bigger magnitude. Formally, the definition of the softmax loss is

$$L_s = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{W_{y_i}^T f_i}}{\sum_{j=1}^{m} e^{W_j^T f_i}} \tag{16}$$

where N is the number of training samples and m is the number of classes. f_i is the feature of ith sample, y_i is the corresponding label in range $[1, n]$, W is the weight matrix before the softmax loss, and W_j is the jth column of W, which is corresponding to the jth class.

In the testing stage, we recognize a sample by

$$\text{Class}(f) = i = \arg\max_j (W_j^T f). \tag{17}$$

In this case, we can infer that $(W_i f - W_j f) \geq 0, \forall j \in [1, n]$. Using this inequality, we obtain the following inference.

For the softmax loss with inner product similarity as its metric, let $P_i(f) = (e^{W_i^T f}/\sum_{j=1}^{n} e^{W_j^T f})$ denotes the probability of x being classified as class i. For any given scale $s \geq 1$, if $i = \arg\max_j (W_j^T f)$, then $P_i(sf) \geq P_i(f)$ always holds. This inference demonstrates that softmax loss always encourages features to have bigger magnitudes. This is the reason why the feature distribution of softmax is "radial."

Due to the "radial" distribution, the magnitude of easy samples becomes much larger than the hard samples. Thus, as shown in Fig. 9, around the zero point, hard samples of each class get mixed up with hard samples of other classes. This property is undesirable in SAR ship recognition. Directly using the features may lead to bad performance.

2) SSC With Softmax Loss: Compared with classic classifiers with softmax loss, the advantages of our SSC can be summarized into four aspects.

1) Instead of processing features in Euclidean space, SSC projects features into spherical space. This is primarily attributed to the fact that spherical features encourage better discriminating performance. For validating the effectiveness of spherical features, the 2-D features in Fig. 9 are normalized to get spherical features, as shown in Fig. 10(a). As a result, the normalized features remove the radial variations and distribute on a sphere, eliminating the effect that the magnitude of easy samples becomes much larger than the hard samples. It can be seen that the spherical features from the same classes are clustered together and those from different classes are pulled apart. Besides, Fig. 10(b) is a diagrammatic sketch of features projected onto a unit hypersphere in high dimension by the SSF layer.
2) Fig. 11(a) and (b) is the structure of MLP and SSC, respectively. After projecting features onto spherical space, we define linear transform, logistic regression, MHSA, GeLU activate function, and other operations on the spherical space under the CNN structure based on the theoretical framework of the proposed SSC. We can leverage the operations defined in the spherical space to fully utilize the properties of the spherical space. Different from classic classifiers (MLPs) process features in Euclidean space, SSC processes features in spherical space (non-Euclidean space). To achieve recognition in spherical space, a series of spherical space layers are further constructed based on the operations defined earlier, such as the SSF layer, SSP layer, and SSR layer.
3) Additionally, SSC combines MHSA to enable the incorporation of long-range feature dependencies into recognition decisions. The MHSA can learn global feature dependencies and realize long-distance feature

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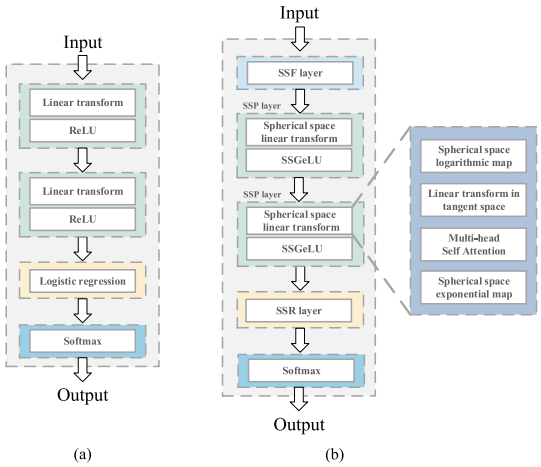


Fig. 11. (a) Structure of MLP. (b) Structure of SSC.

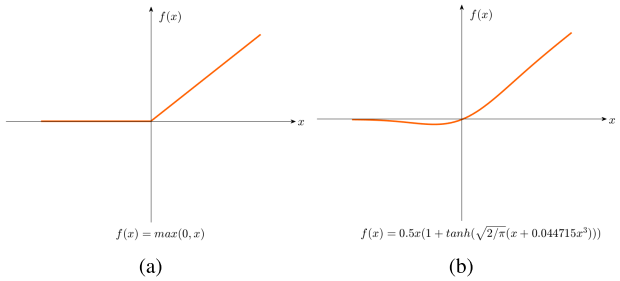


Fig. 12. Comparison of two activation functions. (a) Schematic of ReLU activation function. (b) Schematic of GeLU activation function.

correlation calculation, focusing on the important features and suppressing less significant features.

4) SSGeLU is employed as the activation function to increase the nonlinear relationship between the layers in SSC. As shown in Fig. 12, compared with ReLU, the GeLU has less saturation when the input is close to zero, which means that the output is more sensitive to varieties of the input feature and can better convey subtle differences of the input feature. This is exactly what is very important in SAR ship recognition.

III. Experiments and Results

In this section, we will perform extensive experiments to verify the effectiveness of the proposed method on the OpenSARShip [47] and FUSAR-Ship [48] datasets. First, we describe the two datasets in detail. Then, we present the experimental settings and evaluation indices, which also include the dataset acquisitions, loss function, and dataset settings. Next, the experimental results are demonstrated for OpenSARShip under the three-category recognition task and six-category recognition task identification, and FUSAR-Ship under the seven-category recognition task, respectively. Finally, we perform a comprehensive comparison between the proposed method and some state-of-the-art (SOTA) methods. Ablation studies are also conducted. Last but not least, more discussions of HDSS-Net are present. All the experiments are implemented on a personal computer (PC) with NVIDIA GeForce RTX 2060 VENTUS (12GB) GPU and 24GB RAM. Software development is conducted under the Python language environment using an open-sourced machine learning library PyTorch. CUDA10.1 is used for training and inference acceleration.

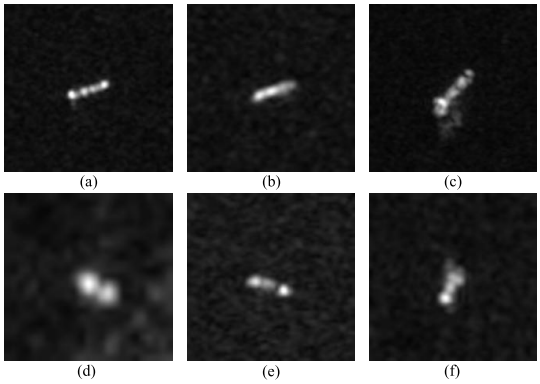


Fig. 13. SAR ship samples in OpenSARShip. (a) Bulk carrier. (b) Cargo. (c) Container ship. (d) Fishing. (e) General cargo. (f) Tanker.

A. Datasets

1) OpenSARShip: Thanks to the release of the OpenSARShip dataset, which presents a promising benchmark for designing advanced algorithms. The biggest character of this dataset is that the labels of ships are based on automatic identification system (AIS) information so the data labeling has high reliability. At present, many researchers have carried out research based on this dataset [28], [30], [49], [50]. OpenSARShip is collected from Sentinel-1, which is a satellite system loaded with C-band SAR [47]. Sentinel-1 has five essential properties, i.e., specifically, large scale, diversity, reliability, and public availability, which exhibit important value in ocean monitoring, sea ice monitoring, marine target detection and identification, emergency monitoring, and humanitarian rescue.

The dataset adopted in this experiment is the ground range detected (GRD) product under Sentinel-1 IW mode with 20m $\times$ 22m resolution and 10.0m $\times$ 10.0m pixel size in azimuth and range directions. Regarding the size of the ship, the length ranges from 92 to 399 m, while the width ranges from 6 to 65 m. Each ship target is represented by pairwise vertical-horizontal (VH) and vertical-vertical (VV) amplitude data samples. It should be known that single-channel VV and VH SAR images are successively inputted to the CNNs. In other words, the CNNs can recognize both VV and VH ships. Fig. 13 presents an example of this dataset.

1) Three-Category: Following Zhang and Zhang [30], we establish a small three-category OpenSARShip dataset to perform experiments, which contains bulk carriers, container ships, and tankers. The three classes of ships account for 80% among the international shipping market [47]. So conducting recognition experiments on this three-category OpenSARShip dataset is meaningful and representative.

In OpenSARShip, the sample numbers of the three-category are imbalanced. To get a category-balanced dataset, the same as HOG-ShipCLSNet [31], the number of training samples is set to be equal for each class. According to the least number

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TABLE I
TRAINING-TEST DIVISION OF THE THREE-CATEGORY DATASET IN OPEN-SARSHIP

Dataset	Category	Train	Test	All
OpenSARShip	Bulk carrier	338	328	666
	Container ship	338	808	1146
	Tanker	338	146	484

TABLE II
TRAINING-TEST DIVISION OF THE SIX-CATEGORY DATASET IN OPEN-SARSHIP

Dataset	Category	Train	Test	All
OpenSARShip	Bulk carrier	200	466	666
	Cargo	200	1142	1342
	Container ship	200	946	1146
	Fishing	200	50	250
	General cargo	200	84	284
	Tanker	200	284	484

of samples in the three-category dataset, the training-testing ratio is set to 7:3. Table I shows the ship sample numbers of the three-category in OpenSARShip.

2) Six-Category: A larger challenging six-category dataset is also established, which contains bulk carriers, cargos, container ships, fishing, general cargos, and tankers. To get a category-balanced dataset, the same as SE-LPN-DPFF [30], the number of training samples is set to be equal for each class. According to the least number of samples in the six-category dataset, the training-testing ratio is set to 8:2. Table II shows the ship sample numbers of the six-category in OpenSARShip.

2) FUSAR-Ship: The high-resolution dataset FUSAR-Ship comes from the Gaofen-3 (GF-3) satellite, which is China's first civil C-band fully polarimetric spaceborne SAR. GF-3 SAR images with azimuth resolution of 1.124 m and slant range resolution ranging from 1.700 to 1.754 m. The FUSAR-Ship dataset is constructed by an automatic SAR-AIS matchup procedure for more than 100 GF-3 scenes, including over 5000 ship image chips with AIS information. FUSAR-Ship is considered an open benchmark dataset for ship and marine target recognition. Fig. 14 shows some SAR ship samples in FUSAR-Ship.

To further confirm the effectiveness of HDSS-Net, a seven-category FUSAR-Ship dataset is established to conduct extensive experiments, which contains bulk carriers, container ships, fishings, tankers, general cargo ships, other cargo ships, and others. Similarly, to get a category-balanced dataset, the same as HOG-ShipCLSNet [31], the training-testing ratio is set to 7:3. Table III shows the ship sample numbers in FUSAR-Ship.

B. Experimental Details and Evaluation Indices

1) Image Preprocessing: The backbone we utilized is pretrained ResNet-50. The pretrained weights of ResNet-50 are based on natural images, which are three-channel images. However, the SAR images we exploit in our article are

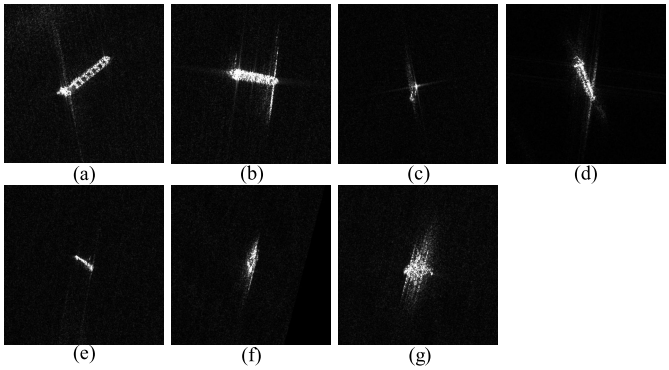


Fig. 14. SAR ship samples in FUSAR-Ship. (a) Bulk carrier. (b) Container ship. (c) Fishing. (d) Tanker. (e) General cargo. (f) Other cargo. (g) Other.

TABLE III
TRAINING-TEST DIVISION OF THE FUSAR-SHIP

Dataset	Category	Train	Test	All
FUSAR-Ship	Bulk carrier	1150	494	1644
	Container ship	1219	523	1742
	Fishing	1101	473	1574
	Tanker	1215	521	1736
	General cargo	1205	517	1722
	Other cargo	1214	521	1735
	Other	1211	520	1731

single channels. In order to utilize the pretrained ResNet-50, we replicate the grayscale value across all three channels. In other words, the values in all three channels are the same at each pixel position since our grayscale image has only one channel.

2) Parameter Setting: These experiments are trained under the same parameters. The size of the input images both in OpenSARShip and FUSAR-Ship are unified to 224 × 224. Using a stochastic gradient descent (SGD) optimizer with the weight decay parameter 0.0005 and the momentum parameter 0.9, the proposed network is trained by 10 000 iterations. The batch size is set to an empirical value of 16 due to the limited GPU memory. To mitigate the negative effects of vanishing training gradients, we set the learning rate as a relatively small value of 0.0005, which is appropriate for our method.

3) Loss Function: The CE loss serves as the loss function of HDSS-Net, defined by

$$L_s = -\frac{1}{N_x}\sum_{\mathrm{i}=1}^{N_x}\log(p_i) \tag{18}$$

where p_i denotes the ith sample recognition result, and N_x denotes the number of all training samples.

4) Evaluation Metrics: For comprehensive evaluations of SAR ship recognition datasets, accuracy (Acc) is used as the core evaluation criterion to measure recognition performance. Acc is the ratio of the number of correctly recognized samples to the total number of samples, which is defined by

$$\mathrm{Acc} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{19}$$

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where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. Furthermore, four other performance metrics are employed for more result verifications in the experiments, including: 1) precision; 2) recall; 3) F1; and 4) confusion matrix.

Precision indicates the number of samples with positive recognitions that are truly positive, which is defined by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{20}$$

Recall indicates the number of positive samples in the input sample that are recognized, which is defined by

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{21}$$

F1 is a combination of both precision and recall, which is defined by

$$\text{F1} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \tag{22}$$

The confusion matrix is an informative table, in which the diagonal entries are the correct recognition sample numbers for each ship category and the nondiagonal entries are the confusions indicating the sample numbers of the ship categories in the rows misclassified into the ship categories in the columns. The sum of each row of the confusion matrix represents the true number of samples in that category, and the sum of each column represents the number of samples recognized to be in that category.

C. Recognition Results

1) Three-Category OpenSARShip Recognition Task: Table IV presents the recognition performance of the proposed method on the OpenSARShip under three-category for each ship class in confusion matrix form. It is observed that the proposed HDSS-Net achieves an accuracy of 83.23%. The diagonal values are significantly higher than others in the same line from Table IV, which indicates that most ships can be recognized correctly. From Table IV, the bulk carrier is recognized as a container ship mistakenly. In fact, as shown in Fig. 13, their class differences are small, and both have functioned as ships that carry goods. Thus much confusion exists.

2) Six-Category OpenSARShip Recognition Task: Table V briefs the recognition performance of the proposed method on the OpenSARShip under six-category for each ship class in confusion matrix form. The SAR ship recognition accuracy achieves 60.76%. We can notice that the accuracy of recognizing the six-category task has decreased dramatically compared with the results of the three-category one. This is mainly because the six-category recognition task is far more challenging than the three-category task. Moreover, the training sample number of the six-category task is fewer than that of the three-category one ($200 < 338$), which further increases the recognition difficulty of the six-category task. Most ships can be correctly recognized, that is, the value of a diagonal is greater than others in the same line. From Table V, cargo, general cargo, and fishing are easily confused, and this may be because the outline of the ship (a strong scattering point) is too vague to provide support for its effective recognition.

3) Seven-Category FUSAR-Ship Recognition Task: The recognition results of the proposed method under seven-category FUSAR-Ship are presented in Table VI. Due to FUSAR-Ship having a better resolution with more ship detail representations that can be learned, the recognition accuracy can reach 90.08%. It is obvious that the confusion of class predictions mainly occurs among fishing, other, and other cargo. It can be expressed as the three ship classes have similar geometries. The bulk carrier, container ship, general cargo, and tanker classes reach nearly a SOTA recognition performance.

D. Comparison With Modern CNN Methods

To comprehensively evaluate the performance improvement of the HDSS-Net, extensive experiments are conducted by comparing the proposed method with classic CNN methods and novel SAR ship recognition CNN methods on OpenSARShip and FUSAR-Ship. Classic CNN methods include AlexNet [13], VGG [14], GoogleLeNet [15], ResNet [16], DenseNet [19], and so on. Novel SAR ship recognition CNN methods include Zeng et al. [27], Xiong et al. [52], SE-LPN-DPFF [30], HOG-ShipCLSNet [31], and so on. Table VII lists the comparison of the quantitative evaluation indices with the CNN methods. The inputs of Zeng et al. [27], Xiong et al. [52], and SE-LPN-DPFF [30] are paired VV-VH SAR amplitude images. The inputs of others are unpaired VV and VH SAR amplitude images, which means single-channel VV and VH SAR images are successively inputted into networks directly.

From Table VII, the following conclusions can be drawn.

1) On the three-category OpenSARShip dataset, HDSS-Net offers the highest recognition accuracy than others. Among all CNN methods, the suboptimal recognition is 79.25% from SE-LPN-DPFF [30]. However, it is still lower than our HDSS-Net by 3.98%, which shows the SOTA SAR ship recognition performance of HDSS-Net.
2) On the six-category OpenSARShip dataset, HDSS-Net also offers the highest recognition accuracy than others. Among all CNN methods, the suboptimal recognition is 56.66% from SE-LPN-DPFF [30]. However, it is still lower than our HDSS-Net by 4.1%. Although SE-LPN-DPFF provides a reasonable solution for dual-polarization SAR ship recognition, it ignores the necessity of utilizing the unique geometric of SAR ship and constructing a powerful task-specific classifier, blocking to reach higher recognition accuracy.
3) On the seven-category FUSAR-Ship dataset, HDSS-Net offers the highest recognition accuracy than others again. The suboptimal recognition is 86.69% from HOG-ShipCLSNet [31], which is still lower than our HDSS-Net by 3.39%. Note that the FUSAR-Ship dataset only provides single-channel SAR images.
4) HDSS-Net obviously improves the recognition accuracy by at least 3% compared with other novel CNN SAR ship recognition methods on both benchmark datasets, which indicates the powerful ability of the proposed methods to extract pretty discriminative representations

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TABLE IV
CONFUSION MATRIX OF HDSS-NET RECOGNITION RESULTS ON THREE-CATEGORY OPENSARSHIP

Dataset	True \ Predicted	Bulk carrier	Container ship	Tanker	**Recall**(%)
OpenSARShip	Bulk carrier	229	91	8	69.81
	Container ship	74	716	18	88.61
	Tanker	5	19	122	83.56
	Precision(%)	73.35	86.68	81.16	**Accuracy=83.23%**
	F1(%)	72.01	87.64	82.99	

TABLE V
CONFUSION MATRIX OF HDSS-NET RECOGNITION RESULTS ON SIX-CATEGORY OPENSARSHIP

Dataset	True \ Predicted	Bulk carrier	Cargo	Container ship	Fishing	General cargo	Tanker	**Recall**(%)
OpenSARShip	Bulk carrier	280	35	138	0	11	2	60.08
	Cargo	193	528	186	0	192	43	46.23
	Container ship	94	13	795	0	40	4	84.04
	Fishing	0	2	1	37	2	8	74
	General cargo	7	22	10	0	41	4	48.81
	Tanker	14	65	18	8	54	125	59.53
	Precision(%)	47.62	79.39	69.25	82.22	15.88	67.21	**Accuracy=60.76%**
	F1(%)	53.13	58.44	75.93	77.89	23.97	53.19	

TABLE VI
CONFUSION MATRIX OF HDSS-NET RECOGNITION RESULTS ON FUSAR-SHIP

Dataset	True \ Predicted	Bulk carrier	Container ship	Fishing	General cargo	Other	Other cargo	Tanker	**Recall**(%)
FUSAR-Ship	Bulk carrier	472	2	2	4	2	8	4	95.55
	Container ship	0	520	1	0	1	1	0	99.43
	Fishing	1	0	411	0	33	27	1	86.89
	General cargo	0	1	0	512	1	0	3	99.03
	Other	0	1	42	0	413	57	7	79.42
	Other cargo	4	0	34	0	54	419	10	80.42
	Tanker	11	0	11	1	12	18	468	90.08
	Precision(%)	96.72	99.24	82.04	99.03	80.04	79.06	94.93	**Accuracy=90.08%**
	F1(%)	96.13	99.33	84.39	99.03	79.73	79.73	92.31	

of SAR ship. On the two benchmark datasets, SE-LPN-DPFF and HOG-ShipCLSNet achieve suboptimal recognition. HOG-ShipCLSNet achieves a preferable recognition result on high-resolution FUSAR-Ship but fails in low-resolution OpenSARShip. This is mainly because HOG-ShipCLSNet needs traditional handcrafted HOG features to assist recognition. However, in low-resolution OpenSARShip, HOG-ShipCLSNet failed to effectively extract traditional handcrafted HOG features due to the low resolution. In contrast, SE-LPN-DPFF provides a reasonable solution for dual-polarization SAR ship recognition, achieving a preferable recognition result on a low-resolution OpenSARShip dataset. However, the variance of SAR image resolution cannot influence the effectiveness of our proposed HDSS-Net. In other words, HDSS-Net can achieve SOTA recognition results for both resolutions.

In conclusion, the designed task-specific network has endowed the proposed ship recognition framework with a great capacity to learn more discriminative representations by utilizing multiscale features and providing multidirection rectangular receptive fields. Moreover, the novel CNN-based SSC can expand the interclass margin and compress the intraclass feature distribution. Hence, the proposed HDSS-Net is well-suitable for ship recognition in both resolution SAR images.

E. Comparisons With Classic Methods for Adjusting Distances Between Classes

The comparisons of some classic loss functions for adjusting distances between classes are also included. Table VIII compares SSC under OpenSARShip and FUSAR-ship with softmax loss, contrastive loss [53], center loss [54], and affinity loss [55] to better demonstrate the effectiveness of the proposed method.

The quantitative experiments show that the proposed SSC achieves higher recognition performance than other methods under OpenSARShip and FUSAR-Ship. On the OpenSARShip dataset, the suboptimal recognition is obtained by center loss. However, it is still lower than SSC by 1.33% on three-category and by 1.78% six-category. On the FUSAR-Ship dataset, the suboptimal recognition is 89.27% from affinity loss, which is still lower than SSC by 0.81%. In conclusion, our proposed SSC achieves the SOTA recognition performance and can further promote the practical applications of SAR ship recognition.

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TABLE VII
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH MODERN CNN METHODS

Model	Three-Category OpenSARShip				Six-Category OpenSARShip				FUSAR-Ship			
	$r(\%)$	$p(\%)$	$f1(\%)$	$Acc(\%)$	$r(\%)$	$p(\%)$	$f1(\%)$	$Acc(\%)$	$r(\%)$	$p(\%)$	$f1(\%)$	$Acc(\%)$
LeNet-5 [12]	65.15	60.54	62.73	65.74	49.14	39.54	43.81	46.59	68.04	68.02	68.03	68.36
AlexNet [13]	68.51	65.52	66.94	70.22	53.40	44.39	48.48	50.98	77.11	77.37	77.24	77.03
VGG-16 [14]	68.61	64.63	66.53	70.05	53.08	45.60	49.06	51.88	80.53	80.67	80.60	80.45
GoogleLeNet [15]	69.73	68.80	69.21	73.80	54.47	44.94	49.25	49.76	83.21	83.37	83.29	82.98
ResNet-50 [16]	71.67	66.79	69.13	72.82	50.27	43.32	46.54	49.80	81.40	81.36	81.38	81.31
Wide-ResNet-50 [18]	73.45	67.64	70.40	73.28	51.31	44.09	47.43	50.73	81.20	81.29	81.25	81.15
DenseNet-161 [19]	72.54	67.77	70.02	73.39	54.98	47.57	51.01	54.27	83.53	83.57	83.55	83.45
MobileNet-v2 [21]	66.01	61.80	63.82	65.83	50.52	43.27	46.61	42.11	74.10	74.10	74.10	74.63
SqueezeNet-v1.0 [23]	71.47	66.73	69.01	72.15	53.24	45.55	49.10	53.12	78.87	79.07	78.97	78.76
Inception-v3 [24]	67.13	63.03	65.00	65.99	54.88	45.15	49.54	50.48	74.70	74.84	74.77	74.61
Inception-v4 [25]	69.26	67.43	68.28	72.44	54.92	46.46	50.34	54.55	80.55	80.89	80.72	80.50
Xception [26]	71.56	68.60	70.00	73.74	52.21	44.03	47.77	49.56	77.42	77.39	77.41	77.29
Wang et al. [32]	57.72	58.72	58.12	69.27	50.53	41.77	45.73	48.43	80.09	80.32	80.21	80.02
Hou et al. [48]	69.93	69.44	66.76	67.41	48.76	41.22	44.67	47.44	71.96	72.19	72.08	71.89
Huang et al. [51]	74.74	69.56	72.04	74.98	54.09	47.58	50.63	54.87	83.13	83.34	83.27	83.19
Zeng et al. [27]	74.99	74.05	74.52	77.41	55.66	47.16	50.96	55.26	-	-	-	-
Xiong et al. [52]	73.87	71.50	72.67	75.44	53.57	45.74	49.35	54.93	-	-	-	-
SE-LPN-DPFF [30]	78.83	76.45	77.62	79.25	54.49	48.61	51.38	56.66	-	-	-	-
HOG-ShipCLSNet [31]	77.87	72.42	75.04	78.15	54.20	46.66	50.15	53.77	86.62	86.54	86.58	86.69
HDSS-Net(Ours)	**80.66**	**81.16**	**80.91**	**83.23**	**59.53**	**60.26**	**59.89**	**60.76**	**90.08**	**90.15**	**90.12**	**90.08**

TABLE VIII
RECOGNITION PERFORMANCE (%) OF CLASSIC METHODS FOR ADJUSTING DISTANCES BETWEEN CLASSES UNDER OPENSAR SHIP AND FUSAR-SHIP

Method	Three-Category OpenSARShip	Six-Category OpenSARShip	FUSAR-Ship
Softmax Loss	77.99	56.17	86.28
Softmax + Contrastive Loss [53]	80.42	58.28	88.65
Softmax + Center Loss [54]	81.90	58.98	88.93
Softmax + Affinity Loss [55]	81.28	57.87	89.27
SSC	**83.23**	**60.76**	**90.08**

TABLE IX
ABLATION STUDY ON FAM

Dataset	FAM	Accuracy(%)
Three-Category OpenSARShip	×	79.72
	✓	**83.23**
Six-Category OpenSARShip	×	57.23
	✓	**60.76**
FUSAR-Ship	×	84.48
	✓	**90.08**

F. Ablation Study

A series of ablation studies on OpenSARShip and FUSAR-Ship datasets are conducted to verify the effectiveness of the FAM, FBM, and SSC in this section.

1) Ablation Study on Feature Aggregation Module: Table IX shows the recognition accuracy with and without the FAM. In Table IX, The "×" means that only the last layer features of the backbone network are utilized, ignoring the multiscale features of middle layers from CNN. The proposed FAM improves the recognition results by 3.51% on three-category OpenSARShip, by 3.53% on six-category OpenSARShip, and by 5.6% on FUSAR-Ship, which is a relatively high improvement. The improvements show the necessity and effectiveness of combining different scale features of CNN to achieve SAR ship recognition of various sizes. Combined with FAM, we create a feature pyramid that has strong representational power at all scales. We can note that FUSAR-Ship achieves the best recognition improvements. The underlying reason may be that high-resolution SAR images can provide more detailed representations to enhance

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TABLE X
ABLATION STUDY ON FBM WITH DIFFERENT KERNEL COMBINATIONS

Dataset	Horizental and Vertical kernels	two Diagonal kernels	Accuracy(%)
Three-Category OpenSARShip	×	×	80.81
	×	✓	82.29
	✓	×	82.52
	✓	✓	**83.23**
Six-Category OpenSARShip	×	×	58.71
	×	✓	59.76
	✓	×	60.26
	✓	✓	**60.76**
FUSAR-Ship	×	×	87.28
	×	✓	89.38
	✓	×	89.83
	✓	✓	**90.08**

the description ability of feature maps. All in all, utilizing multiscale features can improve SAR ship recognition in both resolution SAR images.

2) Ablation Study on FeatureBoost Module: In this section, we seek to investigate FBM through a series of ablation studies. Specifically, we not only focus on verifying the effectiveness of the usage of four rectangular kernels included in FBM as a whole but also divide the four rectangular kernels into two groups and verify the effectiveness of each group. The first group is the horizontal and vertical kernels. The second group is composed of the two diagonal kernels. The "×" means that this group of kernels is not included. When neither of the groups of rectangular kernels is included, only the square kernel is employed. As can be observed in Table X, The proposed FBM improves the recognition results by 2.42% on three-category OpenSARShip, by 2.05% on six-category OpenSARShip, and by 2.8% on FUSAR-Ship, showing rectangular kernels can develop more discriminative SAR ship representations. Only utilizing one group of kernels can also get satisfied improvements by ∼1% on OpenSARShip and ∼2% on FUSAR-Ship. The reason why FBM gets the best performance is mainly because FBM can provide rectangular convolution kernels in all directions, instead of only horizontal and vertical directions or diagonal directions.

We can note that on the base of adding horizontal and vertical kernels, introducing the two diagonal kernels only gains a slight recognition increase. We analyze the cost of the parameters in exchange for recognition improvement. As shown in Table XI, we analyze the number of parameters required by our diagonal kernels. Adding the two diagonal kernels only increases the number of parameters by 0.003M. This constitutes only 6e-5 of the number of parameters of HDSS-Net. In contrast, the diagonal kernels improve the recognition results by 0.71% on three-category OpenSARShip, by 0.50% on six-category OpenSARShip, and by 0.25% on FUSAR-Ship. Although the diagonal kernels are not as impressive as other modules, the added number of parameters of diagonal kernels is so minimal that we can almost ignore them. Thus, introducing the diagonal kernels is worthwhile in exchange for a slight accuracy improvement.

TABLE XI
NUMBER OF PARAMETERS OF HDSS-NET (WITHOUT TWO DIAGONAL KERNELS) AND HDSS-NET

Model	Params(M)
HDSS-Net(w/o two digonal kernels)	48.751
HDSS-Net	48.754

To illustrate the impact of FBM, we display the visualization views of the three-category OpenSARShip and FUSAR-Ship using t-distribution stochastic neighbor embedding (t-SNE) in Figs. 15 and 16, respectively. Given the lower recognition accuracy of the six-category OpenSARShip dataset, the improvements in t-SNE visualization are not remarkable. Therefore, we omit the t-SNE results for the six-category OpenSARShip dataset in this article. In t-SNE, the further the distance and the less the confusion between different classes, the better the recognition accuracy of the model.

Fig. 15(a)–(d) depicts the visualizations for the HDSS-Net (without FBM), the HDSS-Net (with two diagonal kernels), the HDSS-Net (with horizontal and vertical kernels), and HDSS-Net for the three-category OpenSARShip, respectively. Notably, Fig. 15(a) reveals there are many confusions in the feature embedding. Introducing any group of rectangular kernels mitigates recognition errors, resulting in more cohesive feature embeddings for the same class, as displayed in Fig. 15(b) and (c). Subsequent to these improvements, the HSDD-Net demonstrates further enhancement when both groups of rectangular kernels are utilized. Although Fig. 15(b)–(d) shows gradual improvements, we can observe the confusion in the feature embedding among Fig. 15(b)–(d) is similar. This is because their recognition accuracy improvements are not as impressive as other modules.

Fig. 16(a)–(d) illustrates the visualizations for the HDSS-Net (without FBM), the HDSS-Net (with two diagonal kernels), the HDSS-Net (with horizontal and vertical kernels), and the HDSS-Net for the seven-category OpenSARShip, respectively. It is evident that HDSS-Net (with two diagonal

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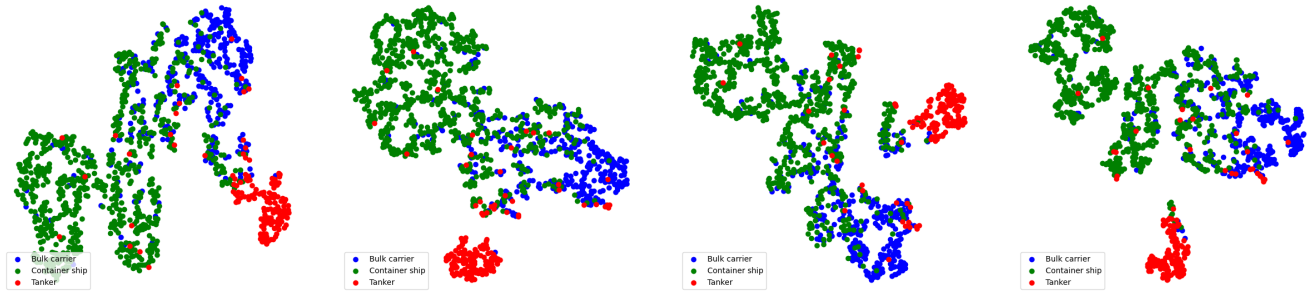


Fig. 15. Ablation experiments on FBM: Three-category OpenSARShip t-SNE feature visualization of the embedding vector distribution. (a) HDSS-Net (without FBM). (b) HDSS-Net (with two diagonal kernels). (c) HDSS-Net (with horizontal and vertical kernels). (d) HDSS-Net.

kernels), HDSS-Net (with horizontal and vertical kernels), and HSDD-Net progressively mitigate recognition errors and enhance the cohesion of feature embeddings within the same class. This observation aligns with the t-SNE visualization of the three-category OpenSARShip.

In conclusion, the designed task-specific convolution kernels break through the limitation of fixed convolution kernels, which show impressive SAR ship recognition improvement in processing elongated SAR ships, and the cost of parameters to exchange for recognition improvement is acceptable.

3) Ablation Study on Spherical Space Classifier: Table XII shows the recognition accuracy comparison with and without the SSC. From Table XII, SSC improves the accuracy by 5.24% on three-category OpenSARShip, by 4.59% on six-category OpenSARShip, and by 3.8% on FUSAR-Ship, respectively. This indicates the effectiveness of the CNN-based SSC. After projecting features onto a sphere and generalizing the basic operations and logistic regression to spherical space, the novel SSC can expand the interclass margin and compress the intraclass feature distribution, which can address the intraclass diversity and interclass similarity.

To further justify the usefulness of SSC, a comprehensive ablation study is conducted to demonstrate that the SSC is more suitable for spherical features. The results are shown in Table XII. HSDD-Net (SSC-Eucl) denotes normalizing learned features to the sphere but the classifier still being Euclidean version. In other words, the linear transform, activation function, and logistic regression in HSDD-Net (SSC-Eucl) are the same as a classic classifier, instead of SSP and SSR. The results show that compared with HSDD-Net (SSC-Eucl), HDSS-Net improves results by 2.38% on three-category OpenSARShip, by 2.08% on six-category OpenSARShip, and by 1.86% on FUSAR-Ship, respectively, demonstrating that SSC is more suitable for spherical feature. HDSS-Net (without exp, log) denotes another way to define the SSP layer by simply normalizing features after each nonlinearity without utilizing spherical exponential and logarithmic maps in SSC. The results show that HDSS-Net (without exp, log) degrades the results of HDSS-Net by 1.09% on three-category OpenSARShip, by 0.87% on six-category OpenSARShip, and by 0.96% on FUSAR-Ship, respectively. This is consistent with our idea that transforming features in spherical space with the exponential and logarithmic maps is reasonable.

To clearly present the effect of the SSC, the qualitative results of three-category OpenSARShip and FUSAR-Ship using t-distribution stochastic neighbor embedding (t-SNE) are shown in Figs. 17 and 18, respectively. Since the recognition accuracy of the six-category OpenSARShip dataset is low, the improvements in the t-SNE visualization are not impressive. So the t-SNE results of the six-category OpenSARShip dataset are not shown in this article.

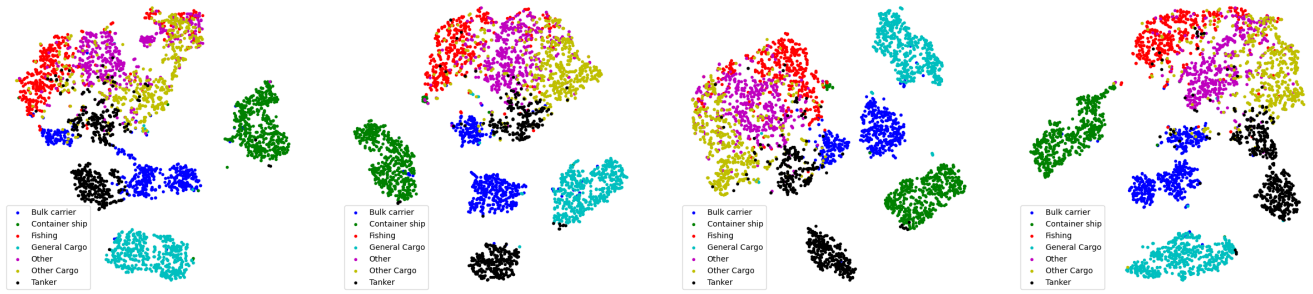
Fig. 17(a)–(d) illustrates the visualization on the HSDD-Net (without SSC), the HSDD-Net (SSC-Eucl), the HDSS-Net (without exp, log), and the HSDD-Net of three-category OpenSARShip, respectively. We can observe that there is much confusion in the feature embedding in Fig. 17(a). The HSDD-Net (SSC-Eucl) alleviates the recognition error and the feature embedding from the same class is more aggregated, which is shown in Fig. 17(b). HDSS-Net (without exp, log) and HSDD-Net show further improvement. However, the confusion between bulk carriers and container ships is still large. This is rational in practice due to both ships function as ships that carry goods, leading to a small class difference.

Fig. 18(a)–(d) illustrates the visualization on the HSDD-Net (without SSC), on the HSDD-Net (SSC-Eucl), the HDSS-Net (without exp, log), and on the HSDD-Net of FUSAR-Ship, respectively. We can see that HSDD-Net (without exp, log) and HSDD-Net can gradually alleviate the recognition error and aggregate the feature embedding from the same class, which is consistent with the t-SNE visualization of three-category OpenSARShip. As shown in Fig. 18, the improvement is particularly pronounced in fishing, other, and other cargo. We can note that in Fig. 18, the feature embedding from the same class has two cluster centers in some classes, such as tanker and bulk carrier. This is mainly because of the defocus in some SAR ship images in the FUSAR-Ship dataset. In conclusion, SSC separates the features between different classes further and the intraclass features of the same class are closer together.

G. More Discussions

1) Recognition Results of the Experiments of the Different Training Samples Under OpenSARShip and FUSAR-Ship: To evaluate the practical application capabilities, recognition experiments are conducted with different numbers of training samples on OpenSARShip and FUSAR-Ship.

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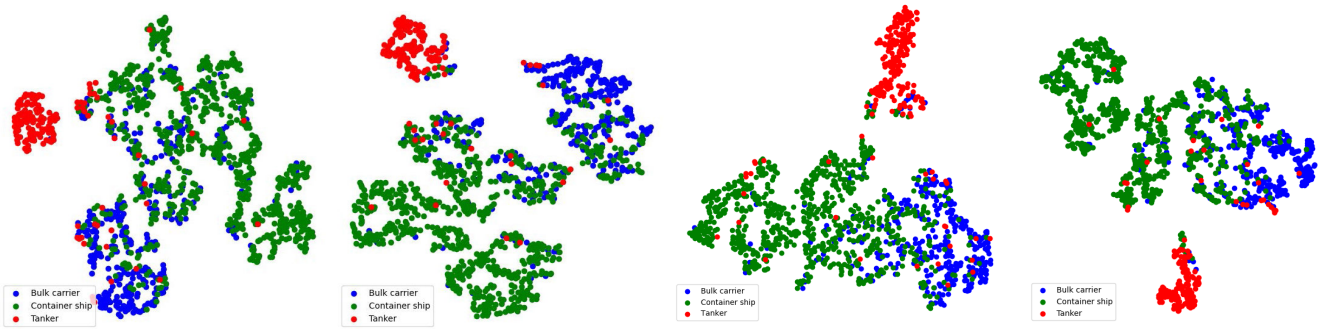


(a) (b) (c) (d)

Fig. 16. Ablation experiments on FBM: FUSAR-Ship t-SNE feature visualization of the embedding vector distribution. (a) HDSS-Net (without FBM). (b) HDSS-Net (with two diagonal kernels). (c) HDSS-Net (with horizontal and vertical kernels). (d) HDSS-Net.

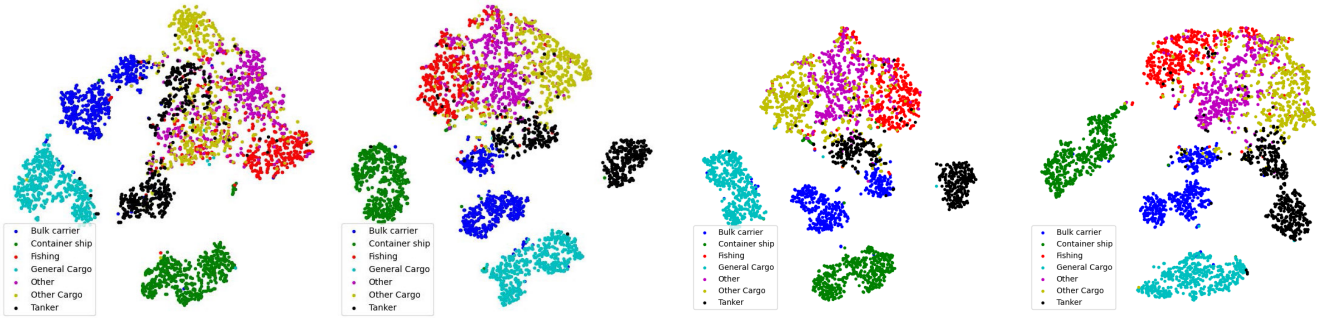
TABLE XII
ABLATION RESULTS (%) ON SSC

Method	Three-Category OpenSARShip	Six-Category OpenSARShip	FUSAR-Ship
HDSS-Net (without SSC)	77.99	56.17	86.28
HDSS-Net (SSC-Eucl)	80.85	58.68	88.22
HDSS-Net (w/o exp, log)	82.14	59.89	89.12
HDSS-Net	**83.23**	**60.76**	**90.08**



(a) (b) (c) (d)

Fig. 17. Ablation experiments on SSC: three-category OpenSARShip t-SNE feature visualization of the embedding vector distribution. (a) HDSS-Net (without SSC). (b) HDSS-Net (SSC-Eucl). (c) HDSS-Net (without exp, log). (d) HDSS-Net.



(a) (b) (c) (d)

Fig. 18. Ablation experiments on SSC: FUSAR-Ship t-SNE feature visualization of the embedding vector distribution. (a) HDSS-Net (without SSC). (b) HDSS-Net (SSC-Eucl). (c) HDSS-Net (without exp, log). (d) HDSS-Net.

1) Recognition Performance of Three-Category OpenSARShip: Table XIII presents the recognition results of the proposed HDSS-Net in three-category OpenSARShip across varying training sample sizes. When the training sample of each class is 338, 100, and 80, respectively, the recognition rates are 83.23%, 79.46%, and 77.46%. It means that when the training sample is enough to train the model, the proposed HDSS-Net attains commendable performance. Notably, even with limited training samples per class, such as 30 or 20, our method achieves

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TABLE XIII
RECOGNITION PERFORMANCE OF EXPERIMENTS OF THE DIFFERENT TRAINING SAMPLES (EDTSS) UNDER THREE-CATEGORY OPENSARSHIP

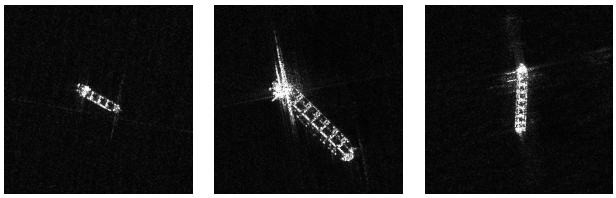
Class	Training Number in Each Class						
	20	30	40	60	80	100	338
Bulk carrier	36.28%	27.74%	69.51%	76.22%	76.22%	69.21%	69.81%
Container ship	82.18%	89.48%	75.62%	75.74%	78.34%	85.52%	88.61%
Tanker	73.97%	66.44%	65.07%	67.12%	75.34%	70.55%	83.56%
Average	69.50%	71.06%	72.85%	74.88%	77.46%	79.64%	83.23%

TABLE XIV
RECOGNITION PERFORMANCE OF EDTSS UNDER SIX-CATEGORY OPENSARSHIP

Class	Training Number in Each Class						
	20	30	40	60	80	100	338
Bulk carrier	58.37%	65.67%	48.07%	72.32%	66.52%	49.14%	60.09%
Container ship	58.67%	57.72%	78.54%	59.41%	72.30%	72.30%	84.04%
Tanker	52.46%	38.38%	28.87%	49.30%	23.94%	52.46%	44.01%
Cargo	45.80%	50.88%	43.43%	52.01%	53.68%	53.59%	46.23%
Fishing	78.00%	68.00%	68.00%	80.00%	80.00%	84.00%	74.00%
General cargo	44.05%	25.00%	41.67%	20.24%	15.48%	39.29%	48.81%
Average	52.99%	53.73%	54.31%	56.86%	58.14%	58.85%	60.76%

TABLE XV
RECOGNITION PERFORMANCE OF EDTSS UNDER FUSAR-SHIP

Class	Training Number in Each Class						
	50	100	150	300	500	800	∼1100
Bulk carrier	57.09%	71.46%	73.68%	81.98%	90.28%	93.72%	95.55%
Container ship	67.50%	78.97%	88.15%	96.75%	97.71%	99.04%	99.43%
Fishing	48.63%	45.45%	52.43%	60.04%	70.40%	80.13%	86.89%
General cargo	64.99%	73.11%	84.33%	93.42%	96.13%	99.03%	99.03%
Other	36.35%	45.00%	40.00%	66.35%	63.65%	77.69%	79.42%
Other cargo	71.40%	69.10%	71.40%	68.91%	75.05%	77.74%	80.42%
Tanker	48.75%	58.16%	60.08%	69.87%	81.00%	84.26%	89.83%
Average	56.49%	63.21%	67.30%	76.94%	82.12%	87.42%	90.08%



(a) (b) (c)

Fig. 19. (a) Bulk carrier1. (b) Bulk carrier2. (c) Bulk carrier3.

accuracy rates of 71.06% and 69.50%, respectively. When the number of training samples is not enough, the proposed method can still achieve a good recognition rate. The achieved recognition performances across this three-category OpenSARShip task underscore the effectiveness of our approach over a broad spectrum of training sample sizes for each class.

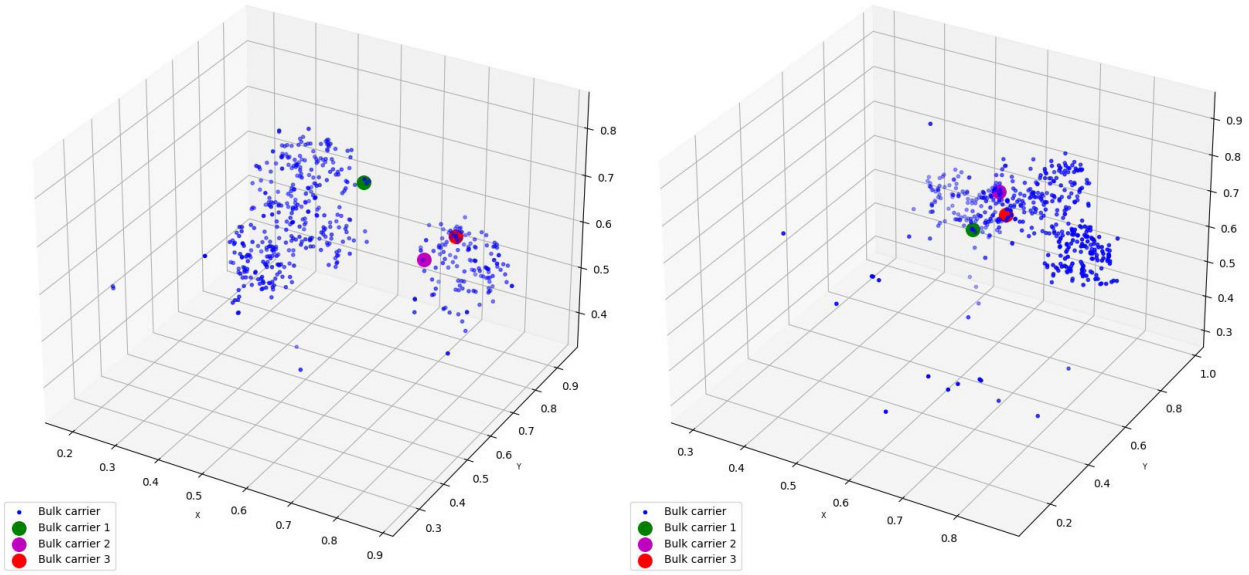
2) Recognition Performance of Six-Category OpenSARShip: The recognition results of HDSS-Net in six-category OpenSARShip across varying training sample sizes are shown in Table XIV. These results show that when training samples are sufficient, the recognition of the HDSS-Net achieves the SOTA, like 60.76% under 338 each class, 58.85% under 100 each class, and 58.14% under 80 each class. When the training samples for each class decrease to 30 or 20, the recognition of SAR ships becomes harder than in other conditions, the proposed HDSS-Net achieves 53.73% and 52.99%. The recognition performance of three-category and six-category over a broad spectrum of training sample sizes for each class on OpenSARShip has validated the effectiveness of our method.

3) Recognition Performance of Seven-Category FUSAR-Ship: The SAR ship recognition performances of seven-category FUSAR-Ship under different training samples are presented in Table XV. It should be noted that different from OpenSARShip, when the number of training samples is insufficient, the recognition performance of FUSAR-Ship decreases obviously. With ample training samples, HDSS-Net achieves impressive performance, such as 90.08% with ∼1100 samples per class, 87.42% with 800 samples per class, and 82.12% with 500 samples per class. However, when the training samples for each class decrease to 100 or 50 the recognition performance only achieves 63.21% and 56.49%. The underlying reason may be compared with OpenSARShip, FUSAR-Ship has more abundant samples and more complex scenarios, such as near-shore regions, islands, and dynamic sea conditions [48], which requires more data to enhance the generalization ability and obtain better recognition results.

2) Spherical Space Feature Visualization at the Individual-Ship Level in One Class: To better illustrate the effectiveness of SSC, we conduct spherical space visualization at the individual ship level in one class. We take the bulk carrier class as an example. In other words, we only visualize the bulk carrier class (the samples are from the FUSAR-Ship dataset). In the bulk carrier class, three samples are selected as the individual-level ship. As shown in Fig. 19, they are various in size and structure, exhibiting intraclass diversity; therefore, these three specific samples are representative.

To present the effect of SSC at the individual-ship level in one class more clearly, we use 3-D t-SNE to present the qualitative results. In the legends of Fig. 20, these three specific samples of bulk carrier class are distinguished from other bulk carrier samples using different symbols for better visual understanding. In Fig. 20, "bulk carrier1" is represented by a green dot, "bulk carrier2" is represented by a purple dot, and "bulk carrier3" is represented by a red dot. The other 'bulk carrier' samples are represented by blue dots.

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(a) (b)

Fig. 20. Visualization of the 3-D feature space using t-SNE at the individual-ship level in the bulk carrier class. (a) Before SSC. (b) After SSC.

TABLE XVI
IMPLEMENTATION COSTS OF DIFFERENT MODELS

Models	Params(M)	FLOPs(G)	Speed/image(ms)
LeNet-5	5.61	0.06	1.5
AlexNet	57.02	0.71	3.1
VGG-16	134.27	15.47	9.9
GoogleLeNet	5.60	1.51	18.9
ResNet-50	23.51	4.13	14.9
ResNext-50-32x4d	22.99	4.29	33.8
Wide-ResNet-50	66.84	11.45	17.9
DenseNet-161	26.48	7.84	60.8
MobileNet-v2	2.23	0.33	15.9
SqueezeNet-v1.0	0.74	0.73	6.9
Inception-v3	21.79	2.85	29.7
Inception-v4	41.15	6.15	48.3
Xception	20.81	4.60	15.9
Wang et al.	15.52	15.35	3.6
Hou et al.	2.75	1.97	8.1
Huang et al.	5.96	2.68	48.9
Zeng et al.	237.03	15.57	9.8
Xiong et al.	0.10	0.86	28.6
SE-LPN-DPFF	4.22	36.53	14.7
HOG-ShipCLSNet	100.23	2.96	31.4
HDSS-Net(Ours)	48.75	5.62	40.9

Fig. 20(a) and (b) illustrates the visualization results before and after the SSC, respectively. In Fig. 20(a), we can observe that the samples of the bulk carrier class are not clustered enough and the distance among the three specific samples is far apart from each other. As shown in Fig. 20(b), after SSC, the samples of the bulk carrier are more aggregated and the three specific samples are getting closer. These qualitative results at the individual-ship level in one class indicate that SSC has a critical impact on the improvement of interclass similarity to achieve better recognition accuracy.

3) Implementation Speed: In addition to the accuracy, we also compare the Params, floating point operations per second (FLOPs), and implementation speed of the proposed HDSS-Net to those SOTA models in Table VII. During the inference, the size of the input image is set to $1 \times 3 \times 224 \times 224$. The results are given in Table XVI. We find

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that the proposed HDSS-Net consumes 40.9 ms for each image. Compared with these methods, the proposed HDSS-Net maintains a certain balance between computational efficiency and recognition accuracy.

4) Disadvantages: First of all, HDSS-Net obtains favorable recognition results at the cost of high model complexity. The number of parameters in our model exceeds that of lightweight CNNs like MobileNet-v2 [21], SE-LPN-DPFF [30], and so on. Second, the proposed method does not consider the imaging mechanism of SAR images. When the SAR images that need to be recognized are defocused, a decline in recognition performance may occur. If the imaging mechanism is introduced in networks, it may refocus the defocused SAR image for further recognition.

5) Practical Applications: Different from Zeng et al. [27], Xiong et al. [52], and SE-LPN-DPFF [30], our approach is not applicable to recognize multipolarized SAR images. Due to the high-resolution dataset FUSAR-Ship only provides single polarimetric SAR data, only single-polarized SAR images are considered in this study. While the utilization of multipolarized SAR images can provide richer information, in situations where data acquisition is restricted in remote areas or computational resources are limited, obtaining single-polarized SAR images might be easier. This is the time HDSS-Net shines through.

6) Future Work: First, we will try to integrate the imaging mechanism and background properties of SAR images into the network in the future, which is a very important issue for SAR applications. In addition, we will try to improve the implementation costs of the model to achieve a better balance between the storage burdens and recognition accuracy in the future. Finally, based on HDSS-Net, we will carry out research on the challenging large-scale ship detection task.

IV. Conclusion

In this article, a task-specific HDSS-Net is proposed for three challenging issues of scale variance, large aspect ratio, and intraclass diversity and interclass similarity, which have restricted the performance of achieving superior SAR ship recognition. The experimental results on the OpenSARShip and FUSAR-Ship datasets show that the proposed HDSS-Net significantly improves the recognition performance on both resolution SAR images. From the ablation study, the FAM, FBM, and SSC have different effects on ship recognition. FAM can obtain a feature pyramid that has strong representational power at all scales, which make the module more suitable for realizing ship recognition with large size variation. FBM breaks the limitation of fixed convolution kernel by providing rectangular receptive fields. The SSC can expand the interclass margin and compress the intraclass feature distribution by fully taking advantage of the property of the spherical space. The three phases of HDSS-Net help each other to achieve optimal recognition performance and tackle the problem of SAR ship recognition. We hope this article will open exciting future directions in the nascent field of geometric deep learning in SAR images.

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Yuanzhe Shang (Graduate Student Member, IEEE) received the B.S. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include radar signal processing, target detection, machine learning, and automatic target recognition.



Wei Pu (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2012 and 2018, respectively.

From 2017 to 2018, he was a Visiting Student with the Department of Electrical Engineering, Columbia University, New York, NY, USA. From 2019 to 2022, he was a Research Fellow with the University College London, London, U.K. He is currently a Professor with the School of Information and Communication Engineering, UESTC. His research interests include sparse signal processing, synthetic aperture radar automatic target recognition, and deep learning.

Dr. Pu was a recipient of the Newton International Fellowship at the Royal Society, U.K.



Congwen Wu (Graduate Student Member, IEEE) received the B.S. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include radar signal processing, machine learning, and automatic target recognition.



Danling Liao received the B.S. degree from the School of Science, Xi'an Jiaotong University, Xi'an, China, in 2020. She is currently pursuing the master's degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China.

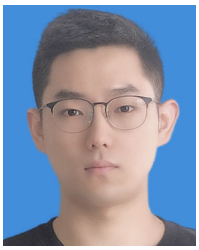
Her research interests include medical image analysis, medical information processing, and computer-assisted interventional diagnosis.

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Xiaowo Xu (Member, IEEE) is currently pursuing the Ph.D. degree in the research group of Xiaoling Zhang with the University of Electronic Science and Technology of China, Chengdu, China.

His research interests include synthetic aperture radar image intelligent interpretation, radar target detection, classification and tracking, deep learning, and computer vision.



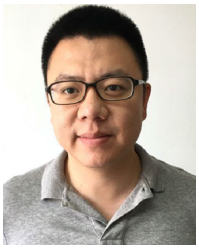
Chenwei Wang (Member, IEEE) received the B.S. degree from the School of Electronic Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2018, where he is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering.

He is currently a Visiting Ph.D. Student with the National University of Singapore, Singapore. His research interests include radar signal processing, machine learning, and automatic target recognition.



Yulin Huang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from the School of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2002 and 2008, respectively.

He is currently a Professor with the UESTC, where he is also the Dean with the School of Information and Communication Engineering. His research interests include radar signal processing and SAR automatic target recognition.



Yin Zhang (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2008 and 2016, respectively.

From 2014 to 2015, he was a Visiting Student with the Department of Electrical and Computer Engineering, University of Delaware, Newark, DE, USA. He is currently a Research Fellow with UESTC. His research interests include signal processing and radar imaging.



Junjie Wu (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2004, 2007, and 2013, respectively.

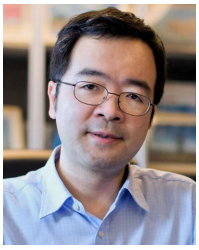
He is currently a Professor with UESTC, where he is also the Vice Dean with the School of Information and Communication Engineering. His research interests include signal processing and radar imaging.



Jianyu Yang (Senior Member, IEEE) received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

From 2001 to 2005, he served as the Dean of the School of Electronic Engineering, UESTC. In 2005, he was a Senior Visiting Scholar with the Massachusetts Institute of Technology, Cambridge, MA, USA. He is currently a Professor with UESTC. His research interests include synthetic aperture radar imaging and automatic target recognition.

Dr. Yang was selected as the Vice Chairperson of the Radar Society of the Chinese Institute of Electronics (CIE) in 2016 and a fellow of CIE in 2018. He is the Vice Chairman of the Radar Society of CIE. He received the titles of the Cross-Century Excellent Talent from the Ministry of Education, the Academic and the Technical Leader from Sichuan Province, and the Top 10 Excellent Scientific and Technological Workers of CIE. He serves as a Senior Editor for the *Chinese Journal of Radio Science* and the *Journal of Systems Engineering and Electronics*.



Jianqi Wu received the B.S. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 1983, and the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China, Chengdu, China, in 1990 and 2018, respectively.

He has been working on radar for more than 30 years. He was in charge of a key national defense advanced research project—Sparse Array Synthetic Impulse and Aperture Radar Experimental System—and several key model projects.

Dr. Wu is a fellow Member of the Chinese Academy of Engineering. He is the Chairperson of the Radar Society of the Chinese Institute of Electronics. He received the First Class Award of the National Scientific and Technological Progress Award, the Second Class Award of the National Scientific and Technological Progress Award three times, the First Class Award of the National Defense Scientific and Technological Progress Prize, and the Outstanding Contribution Award of Science and Technology of Hefei.

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